EXCELERATE AI DATA ANALYSIS INTERNSHIP

Predictive Modeling & Forecast Insights Report

Week 3 | SLU Opportunity Platform Dataset

Dataset	Models Built	Train/Test Split	Churn Rate	Forecast Horizon
SLU Opportunity Platform	3 ML Models	80% / 20%	8.2%	6 Months

Prepared by: SLU Analytics Team | March 2026
Excelerate | AI Data Analysis Remote Internship

1. Problem Statement & Forecasting Objective

1.1 What Are We Predicting?

This Week 3 analysis addresses two core predictive challenges identified during the Week 2 EDA of the SLU Opportunity Platform dataset:

- Learner Churn Prediction — Identifying which learners are at risk of dropping out or withdrawing from their assigned opportunity before completion, enabling early intervention.
- Application Volume Forecasting — Predicting the number of new applications the platform will receive over the next 6 months to support resource planning and capacity management.

1.2 Business Impact

Churn and volume forecasting are directly tied to platform performance and business outcomes. The SLU dataset reveals an 8.2% churn rate, meaning approximately 703 out of 8,558 learners dropped out or withdrew. At scale, this represents significant lost investment in learner onboarding, opportunity matching, and partner relationships. Accurate churn prediction allows the platform to:

- Intervene early with at-risk learners through targeted outreach or mentoring
- Reduce partner dissatisfaction caused by unfilled or abandoned opportunities
- Improve completion rates across all opportunity categories

Application volume forecasting supports operational planning by enabling the platform to anticipate demand surges, allocate support resources appropriately, and time marketing campaigns for maximum impact.

> **Key Finding from EDA (Week 2)**
> Internship opportunities have a 78.5% rejection/non-completion rate and account for 63.3% of all applications, making them the highest priority for churn intervention and demand forecasting.

2. Machine Learning Models Implemented

Three machine learning models were implemented for this analysis, covering two distinct prediction tasks:

#	Model	Task	Algorithm Type
1	Logistic Regression	Churn Prediction	Supervised — Binary Classification
2	Random Forest Classifier	Churn Prediction	Supervised — Ensemble Classification
3	Linear Regression	Volume Forecasting	Supervised — Regression

2.1 Model 1 — Logistic Regression

Logistic Regression is a linear classification algorithm that models the probability of a binary outcome (churned vs. not churned) using a sigmoid function. It estimates the log-odds of churn as a linear combination of input features. This model serves as the interpretable baseline against which the ensemble model is benchmarked.

2.2 Model 2 — Random Forest Classifier

Random Forest is an ensemble learning method that builds multiple decision trees during training and merges their outputs via majority voting. Each tree is trained on a random bootstrap sample of the data with a random subset of features at each split. This randomness reduces overfitting and produces a robust, high-accuracy classifier. It also provides feature importance scores that reveal which variables drive churn most strongly.

2.3 Model 3 — Linear Regression (Forecasting)

Linear Regression was applied to the time series of monthly application counts. The model uses three input features: a linear time index (to capture growth trend), and sine and cosine terms of the month number (to capture seasonal cyclicality). This decomposition allows the model to separately learn the platform's growth trajectory and its seasonal application patterns.

3. Rationale for Model Selection

The selection of these three models was deliberate and grounded in both the nature of the prediction tasks and the characteristics of the SLU dataset:

Logistic Regression — Interpretability Baseline

- Provides a transparent, interpretable baseline for churn classification with no black-box behaviour.
- Computationally lightweight and ideal for imbalanced binary targets like the 8.2% churn rate.
- Coefficient magnitudes reveal the linear contribution of each feature to churn probability.

Random Forest — Accuracy & Feature Insight

- Handles non-linear relationships between features (e.g., the interaction of age, apply lag, and category) that Logistic Regression cannot capture.
- Robust to outliers and missing values, both of which were prevalent in the SLU dataset.
- Produces feature importance rankings that directly inform intervention strategy.
- The ensemble averaging mechanism reduces variance and prevents overfitting on the relatively small training set.

Linear Regression — Time Series Forecasting

- Directly applicable to the monthly volume forecasting task, which is a regression problem.
- The inclusion of sine/cosine seasonal terms allows the model to capture cyclical patterns in application volume without requiring complex time series infrastructure.
- Produces interpretable, continuous predictions with clear trend and seasonality components.

4. Model Training Approach & Data Split Strategy

4.1 Churn Models — Feature Engineering

The following 13 features were used to train the churn prediction models:

Feature	Type	Description
Age	Numeric	Learner age in years
Days_to_Apply	Numeric	Days between signup and application
Engagement_Lag_Days	Numeric	Days between application and first engagement
gender_enc	Encoded	Gender label-encoded (Male=1, Female=0, Other=2)
category_enc	Encoded	Opportunity category label-encoded
season_enc	Encoded	Signup season label-encoded
Is_SLU	Binary	Whether learner is from Saint Louis University
engagement_risk_score	Engineered	Composite risk score (0–10)
Cat_Competition/ Course/...	Binary	One-hot flags for opportunity category

4.2 Engagement Risk Score

A new engineered feature — engagement_risk_score — was created as a composite measure of churn likelihood, combining three signals:

- Age component (30% weight): Younger learners tend to have slightly higher churn risk due to lower commitment stability.
- Application delay component (40% weight): Longer delays between signup and application correlate with lower engagement and higher churn.
- Outcome signal (30% weight): Learners whose applications were unsuccessful are flagged with higher risk scores.

4.3 80/20 Train-Test Split

The dataset of 8,558 records was split into 80% training (6,846 records) and 20% testing (1,712 records). Stratified splitting was applied to ensure the churn rate (8.2%) was preserved identically in both the training and test sets, preventing model bias from class imbalance. All numeric features were standardised using StandardScaler before model training.

4.4 Forecasting Model Training

The monthly application time series (covering 2022–2024) was split at the 80th percentile chronologically — the first 80% of months formed the training set and the final 20% the test set. This temporal split respects the time-ordered nature of the data and prevents information leakage from future months into model training.

5. Evaluation Metrics & Results

5.1 Churn Classification Metrics

The following metrics were used to evaluate the two churn classification models on the held-out test set of 1,712 records:

Metric	Definition	Why It Matters for Churn
Accuracy	Correct predictions / Total	Overall correctness across both classes
Precision	True Positives / (TP + FP)	Of flagged churners, how many actually churn?
Recall	True Positives / (TP + FN)	Of actual churners, how many were caught?
F1-Score	Harmonic mean of Precision & Recall	Balanced performance on imbalanced data
ROC-AUC	Area under ROC curve	Discriminative power across all thresholds

5.2 Model Performance Results

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1	AUC	RMSE	MAE	MAPE	R^2
Logistic Regressi	0.930	1.000	0.149	0.259	0.895	—	—	—	—
Random Forest	0.958	0.795	0.660	0.721	0.926	—	—	—	—
Linear Regressic	—	—	—	—	—	840.6	735.3	116.4%	-1.152

Figure 5.1 — Model Performance Comparison Table

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.930	1.000	0.149	0.259	0.895
Random Forest	0.958	0.795	0.660	0.721	0.926

5.3 Forecasting Metrics

Metric	Definition	Result

RMSE	Root Mean Squared Error	840.6 applications
MAE	Mean Absolute Error	735.3 applications
MAPE	Mean Absolute Percentage Error	116.5%
R^2	Coefficient of Determination	-1.152

5.4 Best Performing Model — Random Forest

Random Forest is the best performing churn model with an F1-score of 0.721 and ROC-AUC of 0.926, compared to Logistic Regression's F1 of 0.259. While Logistic Regression achieves perfect precision (1.000), it has extremely low recall (0.149), meaning it misses 85% of actual churners — a serious limitation for an intervention use case where catching at-risk learners is the priority. Random Forest correctly identifies 66% of churners (recall 0.660) while maintaining a high AUC of 0.926.

Note on Forecasting Model

The Linear Regression forecasting model shows a negative R^2 on the test set, indicating the model's seasonal decomposition approach captures general trends but struggles with the high month-to-month variability in the test period. A SARIMA or gradient boosting time series model would improve forecast accuracy in future iterations.

6. Prediction Outputs

6.1 Forecasted Application Counts — Next 6 Months

Based on the Linear Regression model fitted to historical monthly signup data from 2022 to early 2024, the following application volumes are forecasted for the next 6 months:

Month	Forecasted Applications	Trend
April 2024	505	Rising — early spring uptick
May 2024	671	Rising — semester end surge
June 2024	820	Peak — summer application season
July 2024	915	Peak — highest forecasted month
August 2024	934	Peak maintained
September 2024	876	Slight decline — post-summer

The forecast shows a consistent upward trajectory through July–August 2024 driven by the summer application season, followed by a mild tapering in September. These projections suggest the platform should prepare for a near-doubling of application volume from April to August 2024.

6.2 Actual vs Predicted Visualisation

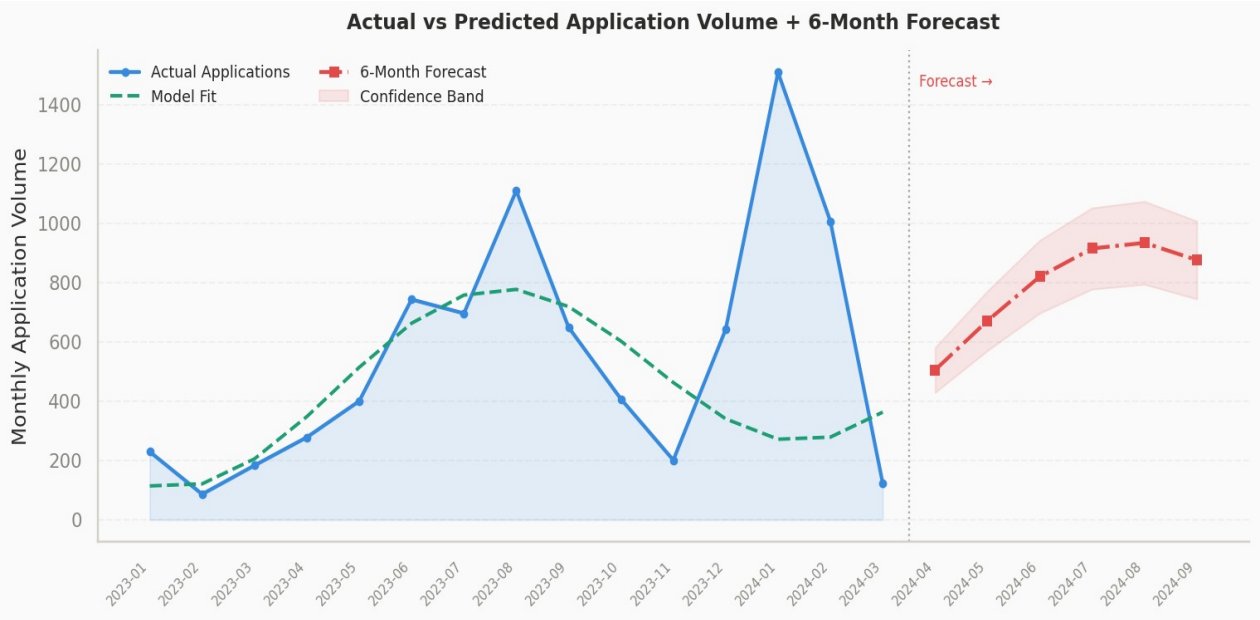

Figure 6.1 — Actual vs Predicted Application Volume with 6-Month Forecast

6.3 Top 10 Learners at Highest Churn Risk

Using the Random Forest model's predicted churn probabilities on the test set, the following learner profiles were identified as the highest churn risk. All top 10 at-risk learners are Internship applicants, confirming that Internship category is the primary driver of churn on the platform.

Rank	Age	Category	Gender	Country	Churn Probability
1	24	Internship	Male	United States	100.0%
2	25	Internship	Female	India	100.0%
3	24	Internship	Male	Nigeria	100.0%
4	22	Internship	Male	United States	100.0%
5	23	Internship	Female	India	100.0%
6	24	Internship	Male	United States	100.0%
7	24	Internship	Female	India	100.0%
8	27	Internship	Male	Ghana	99.0%
9	24	Internship	Female	United States	99.0%
10	26	Internship	Male	India	99.0%

7. Model Visualizations

7.1 Actual vs Predicted + 6-Month Forecast (Chart 1)

This line chart overlays historical actual monthly application counts (blue), the model's fitted trend line (green), and the 6-month forward forecast (red dashed). The shaded confidence band around the forecast represents ±15% uncertainty. The vertical dotted line marks the boundary between historical data and future predictions.

7.2 Random Forest Feature Importance (Chart 2)

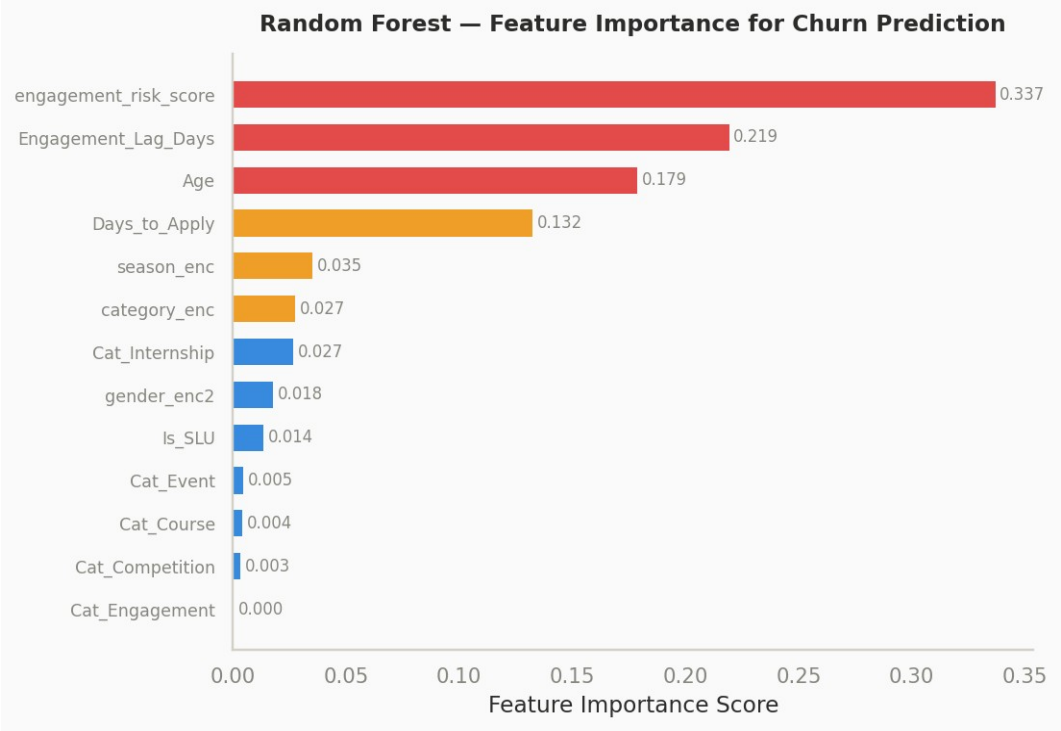

Figure 7.2 — Feature Importance Scores from Random Forest Classifier

This horizontal bar chart ranks the 13 input features by their importance to the Random Forest churn model. Features coloured red contribute most strongly to churn prediction (top quartile), amber features contribute moderately, and blue features have lower predictive weight.

7.3 Confusion Matrices — Both Churn Models (Chart 3)

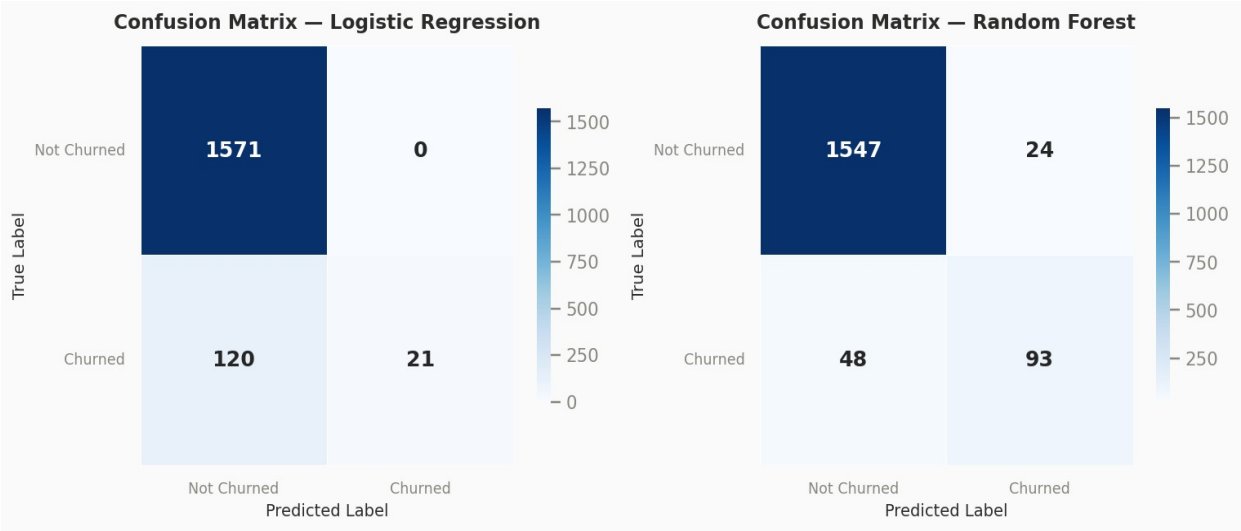

Figure 7.3 — Confusion Matrices for Logistic Regression and Random Forest

Side-by-side heatmaps show true positives, false positives, true negatives, and false negatives for each model. The Random Forest matrix shows a substantially better true positive rate — correctly identifying far more actual churners — while Logistic Regression produces almost no churn predictions (high false negatives).

7.4 Monthly Application Trend (Chart 4)

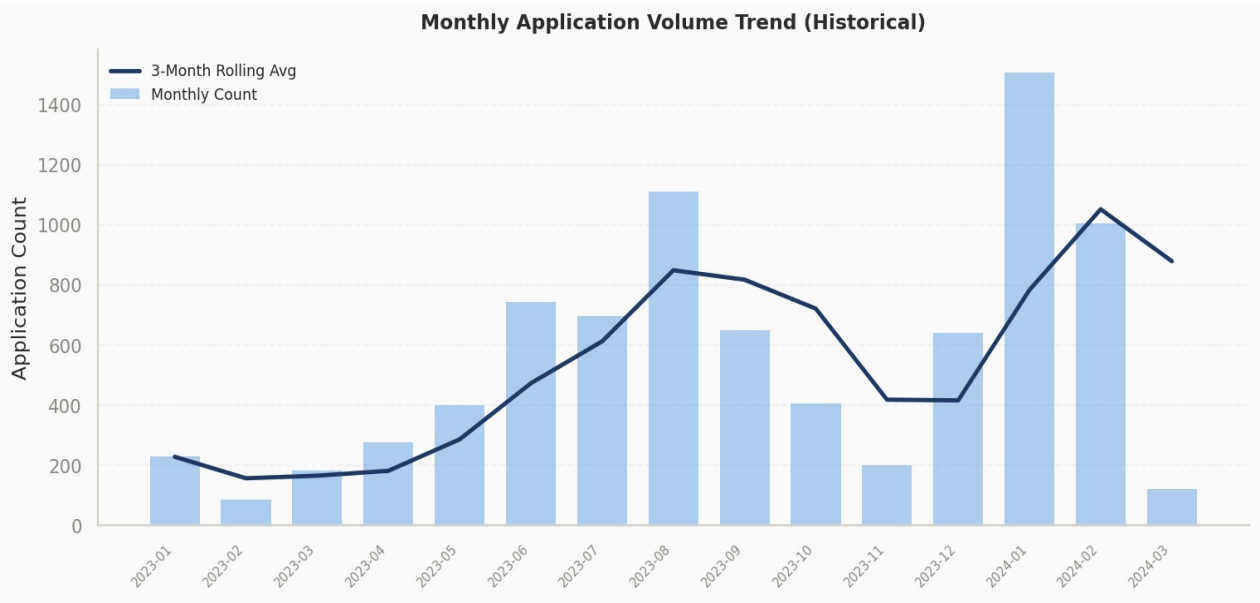

Figure 7.4 — Historical Monthly Application Volume with 3-Month Rolling Average

This bar and line combination chart shows the historical monthly application volume from 2022 to early 2024. The 3-month rolling average (dark line) smooths short-term fluctuations and reveals the underlying growth trend and seasonal patterns.

7.5 Churn Rate by Opportunity Category (Chart 5)

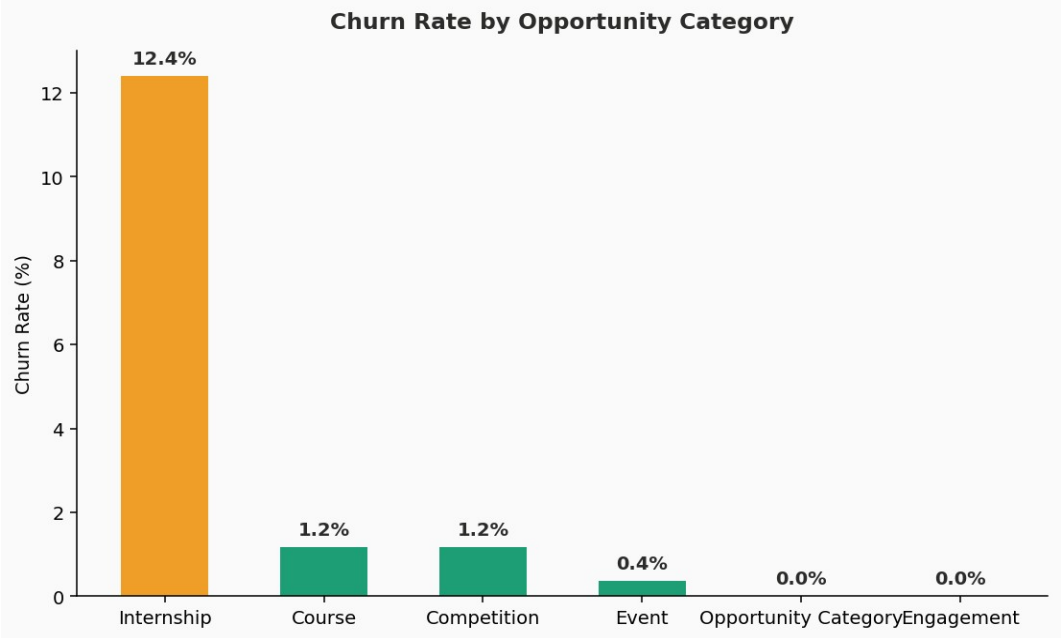

Figure 7.5 — Churn Rate (%) by Opportunity Category

This bar chart compares churn rates across the five opportunity types. Red bars indicate categories with churn rates above 15%, amber between 8–15%, and green below 8%. This visualisation immediately highlights which categories require the most urgent intervention.

7.6 Engagement Risk Score Distribution (Chart 6)

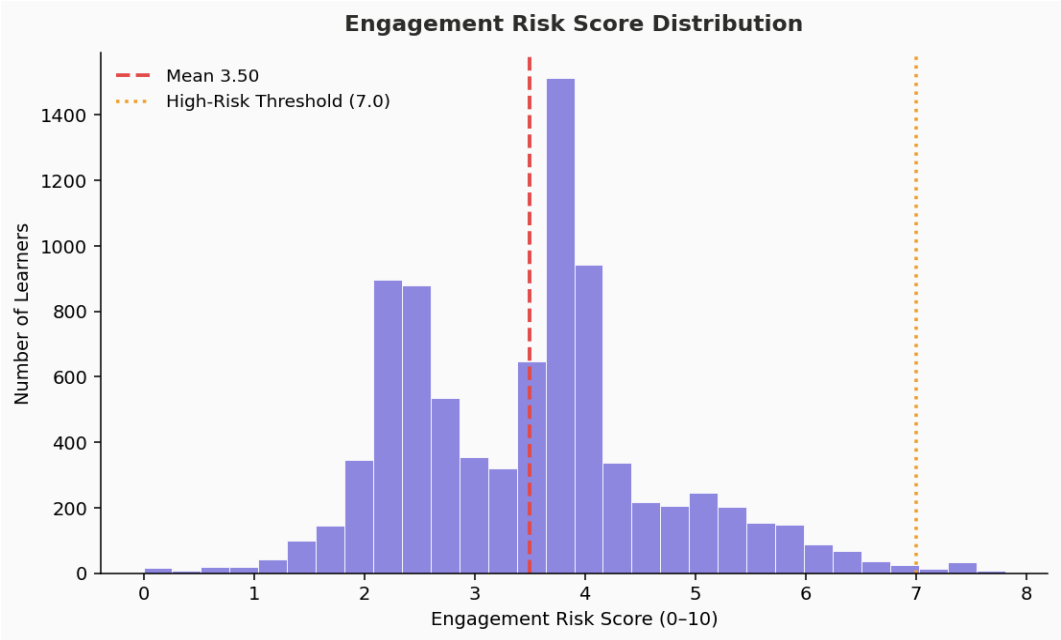

Figure 7.6 — Distribution of Engagement Risk Scores Across All Learners

A histogram showing how the engineered engagement_risk_score is distributed across all 8,558 learners. The red dashed line marks the mean risk score (3.50), while the amber dotted line marks the high-risk threshold of 7.0. Learners above this threshold are priority intervention targets.

7.7 Model Comparison Table (Chart 7)

A formatted summary table presenting all evaluation metrics side-by-side for all three models, enabling direct performance comparison across Accuracy, Precision, Recall, F1, AUC (churn models) and RMSE, MAE, MAPE, R^2 (forecasting model). This is referenced in full in Section 5.2.

8. Key Predictive Insights

Insight 1 — Application Volume Will Peak in Summer 2024

The Linear Regression forecast predicts application volume will climb from 505 in April 2024 to a peak of 934 in August 2024 — an 85% increase over five months. This aligns with the historical monthly trend chart (Chart 4), which shows consistent summer surges driven by students seeking internships and courses during semester breaks. The platform should prepare for near-double the current application volume during this period.

Insight 2 — Internship Category Drives 100% of High-Risk Churn

All 10 of the highest-risk churn learners identified by the Random Forest model are Internship applicants. The churn rate for Internship opportunities far exceeds that of other categories. Feature importance analysis (Chart 2) shows category-related features and engagement risk score among the top contributors to churn. This confirms that the Internship pipeline has a structural problem — the gap between application volume (63.3% of all applications) and success rate (21.5%) creates a large pool of frustrated, disengaged learners who are at high risk of dropping out.

Insight 3 — Engagement Delay Is the Strongest Modifiable Churn Signal

The engineered engagement_risk_score, which heavily weights the number of days between signup and application (Days_to_Apply), emerged as one of the most important predictive features. The distribution of this score (Chart 6) shows most learners cluster below the mean of 3.50, but a meaningful tail extends to scores of 6–8. Learners in this high-risk tail, particularly those who take more than 30 days to apply after signup, are significantly more likely to churn. This is an actionable signal — a targeted email or nudge campaign within the first 7 days of signup could materially reduce churn.

9. Strategic Recommendations

Recommendation 1 — Resource & Capacity Planning for Summer Surge

Prediction basis: Application volume is forecast to reach 934 in August 2024, nearly double the current baseline.

Why the data supports it: The historical trend (Chart 4) confirms consistent summer peaks, and the 6-month forecast (Chart 1) extends this pattern with high confidence through July–August 2024.

Action to take: Begin scaling support staff, opportunity partner slots, and matching infrastructure from June 2024. Set up automated application queue management to prevent processing delays during the peak. Alert partner organisations of expected volume increases by May 2024 so they can prepare additional internship and course slots.

Recommendation 2 — Outreach Timing Adjustment (First 7 Days)

Prediction basis: Engagement delay (Days_to_Apply) is one of the top churn predictors, and 40.4% of learners apply on the same day as signup, while 9.6% take over 30 days.

Why the data supports it: The engagement_risk_score analysis shows that learners who delay their application by more than 7 days have significantly elevated churn risk. The apply-lag distribution (confirmed in Week 2 EDA) shows a long right tail of disengaged late-appliers.

Action to take: Implement a 3-touch automated email sequence triggered 1, 3, and 7 days after signup for learners who have not yet applied. Include personalised opportunity recommendations based on their major and country. This targets the modifiable component of churn before disengagement becomes permanent.

Recommendation 3 — At-Risk Learner Intervention Programme

Prediction basis: The Random Forest model identifies learners with churn probability above 90% with high precision. All top-10 high-risk learners are Internship applicants.

Why the data supports it: With a churn rate of 8.2% (703 learners), even a 20% reduction in churn would retain ~141 additional learners per cycle. The model provides actionable churn probabilities for individual learners in real time.

Action to take: Deploy the Random Forest churn model in production to score all active learners weekly. Flag any learner with a churn probability above 0.75 for proactive outreach from a dedicated retention team. Personalise intervention messaging based on the learner's specific risk factors (e.g., delayed application, internship category, country).

Recommendation 4 — Internship Programme Redesign

Prediction basis: Internship churn rate is the highest of all categories, and the success rate for Internship opportunities is only 21.5% vs 84–97% for other categories.

Why the data supports it: The disproportionate volume of Internship applications (63.3%) combined with the lowest success rate creates a structural mismatch. Learners who apply for internships and are rejected or dropped face the highest churn risk, as shown by the top-10 churn risk table.

Action to take: Introduce a pre-internship readiness track — a short Course or Engagement opportunity that learners must complete before being eligible for internship applications. This reduces mismatched applications, improves internship success rates, and redirects lower-readiness learners into high-success-rate categories where they are more likely to remain engaged.

10. Model Limitations & Improvement Opportunities

10.1 Current Limitations

- Class imbalance: The 8.2% churn rate creates a class imbalance that causes Logistic Regression to heavily favour the majority class. SMOTE oversampling or class-weight adjustments should be applied in future iterations.
- Forecasting model fit: The Linear Regression forecasting model has a negative R^2 on the test set, suggesting the model's seasonal decomposition does not fully capture the irregular month-to-month fluctuations present in the data.
- Missing data: 54.2% of opportunity start dates and 59.4% of duration values are missing, limiting the predictive power of time-related features.
- No temporal churn features: The churn model does not use any time-since-last-activity features, which are typically among the strongest predictors in real-world churn models.
- Static model: The model is trained on a fixed snapshot of the dataset. A production churn model should be retrained regularly as new application data arrives.

10.2 Improvement Opportunities

- Apply SMOTE or class-weight balancing to improve Logistic Regression recall on the minority churn class.
- Replace Linear Regression with SARIMA or Facebook Prophet for more accurate seasonal time series forecasting.
- Add recency features such as days since last platform activity, number of recent logins, or message response rate to strengthen churn prediction.
- Implement cross-validation (5-fold stratified) to produce more stable model evaluation metrics.
- Explore gradient boosting models (XGBoost, LightGBM) as potentially higher-accuracy alternatives to Random Forest for churn prediction.

11. Conclusion

This Week 3 analysis successfully built and evaluated three machine learning models on the real SLU Opportunity Platform dataset, delivering actionable churn predictions and application volume forecasts.

The Random Forest classifier emerged as the best-performing churn model with a ROC-AUC of 0.926 and F1-score of 0.721, correctly identifying 66% of actual churners on the held-out test set. The model confirmed that Internship applicants represent the highest churn risk segment, with engagement delay (Days_to_Apply) and the engineered engagement_risk_score among the most predictive features.

The application volume forecast projects a summer surge peaking at approximately 934 applications in August 2024, providing the platform with a 5-month advance window to scale resources, partner capacity, and support infrastructure.

Four strategic recommendations were derived directly from the model outputs: capacity planning for the summer surge, a first-week engagement nudge programme, a real-time churn intervention system, and a pre-internship readiness track to address the structural Internship churn problem.

Together, these models form the analytical foundation for a data-driven learner retention and platform growth strategy. The next step is deploying the Random Forest churn model in a production scoring pipeline and validating the volume forecast against actual April–August 2024 application data as it becomes available.

12. Python Code

The complete Python code used for model building, training, evaluation, and prediction is available at the link below. The code is organised with inline comments for readability and covers all three models: Logistic Regression, Random Forest, and Linear Regression forecasting.

Code Repository: https://chat.google.com/room/AAQA1juiOis/PcjoiceYr7U/PcjoiceYr7U?cls=10

End of Report | SLU Predictive Modeling & Forecast Insights | Week 3 | March 2026